Equality Cases in the Sharp One-Dimensional Matrix Lieb–Thirring Inequality

Quantitative Potential Reconstruction and Conditional Channel Stability

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M525

October 9, 2026

Abstract

The source classifies equality potentials by a constant unitary decomposition into scalar one-state profiles. Its action method retains a first-order residual, a truncated algebraic defect, and a matrix Young deficit. We derive a quantitative global L^p reconstruction bound for the physical potential directly from the Young deficit and the source coercive estimate. On rank-stable finite intervals we convert the truncated action defect into explicit bounds on left and physical right channel rotation, including energy-gap costs and the entire bad set. We also give a local Riccati residual estimate and an explicit trial-Gram error budget. Rotating rank-one examples explain why rank floors and spectral separation cannot be omitted. Form domains, matrix order, noncommutation, common spectral shifts and the physical metric are preserved. The source's global path construction, infinite-row compactness and equality classification remain distinct from these local and conditional deductions. Contribution assessment: C3/5, quantitative physical-potential reconstruction and defect-to-channel stability refine the source equality mechanism with explicit rank and gap costs, without an independent global classification or near-extremizer theorem.

1 The operator and the source theorem

Fix a finite physical dimension $m \geq 1$ and

$$\tfrac{1}{2} < \gamma < \tfrac{3}{2}, \quad \sigma = \gamma - \tfrac{1}{2} \in (0,1), \quad p = 1 + \sigma, \quad r = \sigma^{-1} > 1, \quad q = r + 1.$$

The physical Hilbert space is $L^2(\mathbb{R}; \mathbb{C}^m)$, with inner product conjugate linear in the first argument. Let $W(x) = W(x)^* \geq 0$ be measurable and $I_W = \int_{\mathbb{R}} \operatorname{tr}(W^p) < \infty$. All finite matrix traces are unnormalized. The operator H_W is defined by the closed form

$$h_W[u,v] = \int_{\mathbb{R}} \langle u', v' \rangle - \int_{\mathbb{R}} \langle u, Wv \rangle, \qquad D(h_W) = H^1(\mathbb{R}; \mathbb{C}^m). \tag{1}$$

There is no boundary at a finite endpoint and no imposed gauge. With $w = \|W\|_{\mathrm{op}} \in L^p$, one-dimensional Sobolev, interpolation and Young inequalities give

$$\int \langle u, Wu \rangle \leq 2^{1/p} \|w\|_p \left\|u'\right\|_2^{1/p} \|u\|_2^{2-1/p} \leq \varepsilon \left\|u'\right\|_2^2 + C_\varepsilon \|u\|_2^2 .$$

Thus a sufficiently shifted form norm is equivalent to H^1; the associated operator is self-adjoint and bounded below. Its exact domain is

$$D(H_W) = \{u \in H^1 : \ -u'' - Wu \in L^2 \text{ in the distributional sense}\}. \tag{2}$$

Indeed the form representation theorem characterizes the operator by $h_W[v,u] = \langle v, f\rangle$ for every $v \in H^1$. Testing against compactly supported smooth v gives the stated distribution; the converse extends that identity by H^1 density and continuity of the form.

The source proves compactness of $W^{1/2} : H^1 \to L^2$ by truncating both position and $\|W\|_{\mathrm{op}}$, then applying compact local Sobolev embedding. Consequently negative eigenvalues have finite multiplicity and can accumulate only at zero. This form argument does not assert that every smooth trial function is in $D(H_W)$.

Example 1.1 (Form vectors need not have L^2 operator residuals)**.** Take $p = 3/2$, $m = 1$, and $W(x) = |x|^{-7/12}\mathbf{1}_{\{0<|x|<1\}}$. Then $W \in L^{3/2}$ since $7/8 < 1$, but $W \notin L^2$ since $7/6 > 1$. A smooth compactly supported function equal to one near zero belongs to the form domain but not the operator domain: its second derivative vanishes near zero while $Wu \notin L^2$. Hence a trial-form calculation is legitimate although its alleged L^2 differential residual is not. Distributional or form-dual residuals must not be reported as L^2 residuals.

Put

$$D_\sigma = \frac{\sigma^\sigma}{(1+\sigma)^{1+\sigma}}, \qquad b_\sigma = \int_{-1}^{1} (1-t^2)^\sigma \, dt, \qquad C_\gamma = D_\sigma / b_\sigma.$$

The source [1] states the sharp bound $\mathrm{Tr}(H_W)_-^\gamma \le C_\gamma I_W$, and classifies equality as

$$W(x) = U\left[\mathrm{diag}\big((r+1)a_j^2 \operatorname{sech}^2(r a_j(x-x_j))\big)_{j=1}^{k} \oplus 0_{m-k}\right] U^*, \tag{3}$$

U constant unitary, $0 \le k \le m$, $a_j > 0$ and $x_j \in \mathbb{R}$. Repeated scales and identical profiles are allowed. The formulation excludes position-dependent unitary diagonalization. Each listed nonzero scalar channel has one negative eigenvalue, $-a_j^2$. We attribute this complete classification and its general sharp inequality to the source; the finite certificates below are not an independent proof of either global assertion.

2 Young deficit gives quantitative potential reconstruction

For positive semidefinite m-by-m matrices W, Z, define

$$Y(W,Z) = D_\sigma \,\mathrm{tr}(W^p) - \mathrm{tr}(WZ) + \mathrm{tr}(Z^q), \qquad W_0 = qZ^r. \tag{4}$$

The source proves $Y \ge 0$, with equality exactly when $W = W_0$, and the useful coercive estimate

$$\mathrm{tr}(Z^q) \le 2Y(W,Z) + 2(2^{p-1}-1)D_\sigma \,\mathrm{tr}(W^p). \tag{5}$$

We retain this valid source estimate and extract a quantitative conclusion that does not require a uniform pointwise cap.

Lemma 2.1 (Noncommutative quadratic lower bound)**.** *Let $M = \max(\|W\|_{\mathrm{op}}, \|W_0\|_{\mathrm{op}}) > 0$. Then*

$$Y(W,Z) \ge \frac{D_\sigma p(p-1)}{2} M^{p-2} \|W - W_0\|_{\mathrm{HS}}^2 . \tag{6}$$

If $M = 0$, both sides are interpreted as zero.

Proof. The identities $D_\sigma p W_0^{p-1} = Z$ and $D_\sigma(p-1)\operatorname{tr}(W_0^p) = \operatorname{tr}(Z^q)$ show that Y is D_σ times the trace Bregman difference for t^p. Set $c = p(p-1)M^{p-2}$ and $g(t) = t^p - ct^2/2$ on $[0, M]$. It is convex because $p < 2$ and $g''(t) = p(p-1)t^{p-2} - c \geq 0$ for $t > 0$; it is continuously differentiable at zero. For Hermitian X, Y_1 with spectra in this interval,

$$\operatorname{tr} g(X) \geq \operatorname{tr} g(Y_1) + \operatorname{tr}(g'(Y_1)(X - Y_1)).$$

To prove this without any commutation assumption, diagonalize Y_1. In each of its basis vectors, scalar Jensen applied to the spectral measure of X, followed by the scalar tangent inequality at the corresponding eigenvalue of Y_1, gives the assertion after summation. Apply it to $X = W, Y_1 = W_0$ and move the quadratic terms to the right. Their trace is $c\operatorname{tr}((W - W_0)^2)/2$, proving (6), including zero eigenvalues. □

Theorem 2.2 (Global physical L^p reconstruction)**.** *Let W, Z be measurable positive semidefinite matrix fields on $\mathbb{R}$, $I_W < \infty$, and $E_Y = \int_{\mathbb{R}} Y(W, Z) < \infty$. Define*

$$J = [1 + 2q^p(2^{p-1} - 1)D_\sigma]I_W + 2q^p E_Y.$$

Then

$$\|W - qZ^r\|_{L^p(\mathrm{HS})} \leq \left(\frac{2E_Y}{D_\sigma p(p-1)}\right)^{1/2} J^{(2-p)/(2p)}. \tag{7}$$

The matrix size does not occur as an additional factor.

Proof. Pointwise,

$$M^p \leq \|W\|_{\mathrm{op}}^p + q^p \|Z\|_{\mathrm{op}}^{rp} \leq \operatorname{tr}(W^p) + q^p \operatorname{tr}(Z^q).$$

Here $rp = q$. Integration and (5) give $\int M^p \leq J$. Raise (6) to $p/2$ after solving for the squared Hilbert–Schmidt difference. Hölder with conjugate exponents $2/p$ and $2/(2-p)$ yields

$$\int \|W - qZ^r\|_{\mathrm{HS}}^p \leq \left(\frac{2}{D_\sigma p(p-1)}\right)^{p/2} E_Y^{p/2} \left(\int M^p\right)^{(2-p)/2}.$$

Taking the pth root proves the theorem. Points with $M = 0$ contribute zero; no inverse at a zero eigenvalue was used. □

For the source equality sequence, $Z_j = A_j^* A_j$ and $E_{Y,j} \leq d_j \to 0$. The theorem therefore gives the explicit strong convergence $qZ_j^r \to W$ in the physical $L^p(\mathbb{R}; \mathrm{HS})$ norm, with the displayed $O(\sqrt{d_j})$ bound when I_W is fixed and $d_j \leq 1$. This strengthens this particular potential-identification step. It does not show that qZ_j^r is a soliton potential or that a near-extremizer lies near the global soliton family. Classical trace convexity and Hölder are the tools; the source-specific increment is their combination with its actual Young/coercivity budget.

3 Projection derivatives with the physical metric

Let $I = [a, b]$ be a finite interval. Suppose $A \in H^1(I; \mathbb{C}^{n\times m})$ has constant rank $k > 0$ and all its positive singular values are at least $s > 0$. Write $A^\dagger$ for the Moore–Penrose inverse and

$$P = AA^\dagger, \qquad Q = A^\dagger A.$$

Thus P acts on the row-index space $\mathbb{C}^n$, whereas Q acts on the physical space $\mathbb{C}^m$. They are different projectors. The zero-rank case has constant zero projectors and needs no inverse.

Lemma 3.1 (Off-diagonal derivative identity)**.** *The projectors are absolutely continuous and, almost everywhere,*

$$P' = Z_L + Z_L^*, \qquad Z_L = (I-P)A'A^\dagger, \tag{8}$$

$$Q' = Z_R + Z_R^*, \qquad Z_R = A^\dagger A'(I-Q). \tag{9}$$

Moreover $\|P'\|_{\mathrm{op}} = \|Z_L\|_{\mathrm{op}}$ *and* $\|Q'\|_{\mathrm{op}} = \|Z_R\|_{\mathrm{op}}$.

Proof. The positive spectral gap at zero makes the range projection of AA^* locally a contour integral of its resolvent. Since A is absolutely continuous and bounded on I, differentiating that resolvent proves absolute continuity of P; the same applies to A^*A and Q. One may use finitely many such neighborhoods on the compact interval. Differentiating $PA = A$ gives $P'P = (I-P)A'A^\dagger$. Differentiating $P^2 = P = P^*$ shows that P' has zero diagonal blocks relative to P, giving (8). The proof for Q is the adjoint argument. An off-diagonal Hermitian block matrix $\left(\begin{smallmatrix} 0 & Z \\ Z^* & 0 \end{smallmatrix}\right)$ has operator norm $\|Z\|_{\mathrm{op}}$, by squaring it. Thus no factor two is required in these operator-norm estimates. □

4 The actual truncated action defect controls drift

Let $K = K^* > 0$ be constant, $\|K\|_{\mathrm{op}} \le \kappa$, and let $B(x) = B(x)^*$ be measurable with $\|B\|_{\mathrm{op}} \le \kappa$. Set

$$R = AA^*, \quad D = A' - BA, \quad F = K^2 - B^2 - R^r, \quad C = [B, K^2], \quad \Phi = \min\{1, \|C\|_{\mathrm{op}}^2 + \|F\|_{\mathrm{op}}^2\}.$$

These are precisely the algebraic quantities in the source action estimate (with adjoints for complex matrices). Fix a constant $c_* > 0$ and a certified budget

$$\int_I (\|D\|_{\mathrm{HS}}^2 + c_*\Phi) \le \mathcal{E}. \tag{10}$$

In a source application, obtaining this budget with its prescribed path and boundary values is an upstream obligation. Arbitrary small sampled residuals do not supply it.

Theorem 4.1 (All-set finite-interval channel drift)**.** *Under these hypotheses and the rank floor of Section 3, define*

$$C_s = \max\{\kappa, \sqrt{1+4\kappa^2}\, s^{-2r}\}.$$

Almost everywhere,

$$\|P'\|_{\mathrm{op}} \le s^{-1}\|D\|_{\mathrm{HS}} + C_s\sqrt{\Phi}, \tag{11}$$

$$\|Q'\|_{\mathrm{op}} \le s^{-1}\|D\|_{\mathrm{HS}}. \tag{12}$$

Consequently for $x, y \in I$,

$$\|P(x) - P(y)\|_{\mathrm{op}} \le \min\{1, \sqrt{|x-y|(s^{-2} + C_s^2/c_*)\mathcal{E}}\}, \tag{13}$$

$$\|Q(x) - Q(y)\|_{\mathrm{op}} \le \min\{1, s^{-1}\sqrt{|x-y|\mathcal{E}}\}. \tag{14}$$

All constants are independent of the number of rows n.

Proof. Since R^r has range P and is at least s^{2r} there,

$$(I-P)[B,R^r]P=(I-P)BPR^r,\qquad [B,R^r]=C-[B,F].$$

It follows that

$$\|(I-P)BP\|_{\mathrm{op}}\le s^{-2r}(\|C\|_{\mathrm{op}}+2\kappa\,\|F\|_{\mathrm{op}}).$$

On the set where $\|C\|^2+\|F\|^2<1$, this is at most $\sqrt{1+4\kappa^2}s^{-2r}\sqrt{\Phi}$. On its complement, $\Phi=1$ and the independent bound $\|(I-P)BP\|_{\mathrm{op}}\le\kappa$ applies. This is why the maximum in C_s is present: the bad set is not discarded. Insert $A'=BA+D$ into (8) and use $\left\|A^\dagger\right\|_{\mathrm{op}}\le s^{-1}$. For (9), the term $BA(I-Q)$ vanishes exactly. These prove the derivative bounds.

Integration along the interval from x to y, followed by Cauchy–Schwarz applied jointly to $(\|D\|_{\mathrm{HS}},\sqrt{c_*\Phi})$, gives the square-root constants. The norm of the difference of two orthogonal projections is at most one. This proves both integrated estimates. □

Corollary 4.2 (Physical channels of an isolated energy block)**.** *Let* Π *project onto the eigenspace* $K^2=\alpha^2$*, and suppose its distance to the rest of the spectrum of* K^2 *is* $g>0$*. Assume* $\|A\|_{\mathrm{op}}\le M$ *on* I*, and* $G=\Pi A$ *has constant positive rank with positive singular values at least* $s_\alpha>0$*. Let* $Q_\alpha=G^\dagger G$*. With* $C_\kappa=\max\{1,2\kappa^3\}$*,*

$$\|Q_\alpha(x)-Q_\alpha(y)\|_{\mathrm{op}}\le\min\left\{1,\frac{\sqrt{|x-y|\mathcal{E}}}{s_\alpha}\sqrt{1+\frac{M^2C_\kappa^2}{g^2c_*}}\right\}. \tag{15}$$

If $\Pi=I$*, the second term inside the square root is omitted.*

Proof. The exact block identity

$$\Pi B(I-\Pi)=\Pi C(I-\Pi)(K^2|_{\mathrm{ran}(I-\Pi)}-\alpha^2 I)^{-1}$$

gives $\|\Pi B(I-\Pi)\|_{\mathrm{op}}\le\|C\|_{\mathrm{op}}/g$. Here the block is a single repeated eigenvalue; no simultaneous diagonalization of B and K is assumed. Thus $G'=(\Pi B\Pi)G+D_\alpha$, where $\|D_\alpha\|_{\mathrm{op}}\le\|D\|_{\mathrm{HS}}+M\,\|C\|_{\mathrm{op}}/g$. On the good set $\|C\|_{\mathrm{op}}\le\sqrt{\Phi}$; on the bad set $\|C\|_{\mathrm{op}}\le 2\kappa^3$. Hence $\|C\|_{\mathrm{op}}\le C_\kappa\sqrt{\Phi}$ everywhere. The right-projector proof of (12), with operator norm for this rectangular residual, gives $\|Q'_\alpha\|_{\mathrm{op}}\le\|D_\alpha\|_{\mathrm{op}}/s_\alpha$. Joint Cauchy–Schwarz proves the result. □

The source uses one common downward shift of the squared binding energies. This preserves every difference of their squares and hence preserves the gap g in this corollary. Different shifts chosen independently would not preserve that useful property. Even these bounds do not control the rotation *within* a full-rank physical block; then $Q_\alpha=I$ carries no such information. The Riccati and algebraic relations have a separate role.

5 A decaying-kernel Riccati residual bound

On an exact active source block the logarithmic derivative satisfies $B'=r(B^2-a^2I)$, with $-aI<B<aI$. A generic Gronwall estimate for a perturbed equation grows exponentially with interval length. The following classical Cayley linearization instead gives an explicit stable forward estimate, at the price of a declared distance from one spectral endpoint.

Theorem 5.1 (Forced Cayley equation)**.** *Let $a, r > 0$, let $B \in W^{1,1}([x_0, x_1]; \mathbb{C}^{k\times k})$ be Hermitian, and assume $-aI \le B(x) < aI$ throughout the interval. Put*

$$\mathcal{F} = B' - r(B^2 - a^2 I), \quad T = (aI + B)(aI - B)^{-1}, \quad T_0(x) = e^{-2ra(x-x_0)}T(x_0),$$

and $B_0(x) = aI - 2a(I + T_0(x))^{-1}$. Then $T, T_0 \ge 0$, B_0 is the unforced Riccati solution with the same initial matrix, and

$$T' + 2raT = 2a(aI - B)^{-1}\mathcal{F}(aI - B)^{-1} \quad \text{almost everywhere.} \tag{16}$$

Writing $\eta(x) = a - \lambda_{\max}(B(x)) > 0$, we have

$$\|B(x) - B_0(x)\|_{\mathrm{op}} \le 4a^2 \int_{x_0}^{x} e^{-2ra(x-t)} \frac{\|\mathcal{F}(t)\|_{\mathrm{op}}}{\eta(t)^2}\, dt. \tag{17}$$

In particular, if $\eta \ge \eta_0 > 0$ and $\|\mathcal{F}\|_{\mathrm{op}} \le f$, the right side is at most

$$\frac{2af}{r\eta_0^2}(1 - e^{-2ra(x-x_0)}).$$

Proof. Since $T = 2a(aI - B)^{-1} - I$, the derivative of the inverse gives

$$T' = 2a(aI - B)^{-1}B'(aI - B)^{-1}.$$

Only the polynomial part of B' commutes with B. Its contribution is $-2raT$; the forcing remains between the two inverses in exactly the order shown in (16). Variation of constants yields the decaying-kernel integral for $T - T_0$. Spectral calculus gives $T \ge 0$ and $T_0 \ge 0$. The inverse identity then implies

$$\|B - B_0\|_{\mathrm{op}} = 2a \left\|(I + T)^{-1}(T - T_0)(I + T_0)^{-1}\right\|_{\mathrm{op}} \le 2a \|T - T_0\|_{\mathrm{op}}.$$

Together with $\left\|(aI - B)^{-1}\right\|_{\mathrm{op}} = \eta^{-1}$ this proves the bounds. Direct differentiation verifies the unforced equation for B_0. All differentiations are valid in $W^{1,1}$: continuity and strict positivity on the compact interval keep the inverse bounded. □

If the initial spectrum is strictly inside $(-a, a)$, diagonalizing $T(x_0)$ by one constant unitary gives eigenvalues $e^{2ra(y_j - x_0)} > 0$, and hence

$$B_0(x) = U \operatorname{diag}(-a \tanh(ra(x - y_j)))U^*.$$

This is the source profile mechanism, not a new inverse-scattering classification. The chart becomes ill-conditioned as $\lambda_{\max}(B) \uparrow a$; for an exact soliton this occurs towards the left infinite tail. Replacing x by $-x$ and B by $-B$ gives the corresponding reverse estimate controlled by the distance from $-a$. Neither chart is asserted uniformly well-conditioned on the whole line.

Corollary 5.2 (How second-order and algebraic errors enter)**.** *Let D be a C^2 invertible square matrix field, with $B = D'D^{-1}$ Hermitian and satisfying the hypotheses of Theorem 5.1. Define*

$$E_{\mathrm{alg}} = a^2 I - B^2 - (DD^*)^r, \qquad S = D'' - a^2 D + qD(D^*D)^r, \qquad q = r + 1.$$

Then

$$\mathcal{F} = qE_{\mathrm{alg}} + SD^{-1}. \tag{18}$$

If the smallest singular value of D is at least $s_D > 0$, one may insert $q\|E_{\mathrm{alg}}\|_{\mathrm{op}} + s_D^{-1}\|S\|_{\mathrm{op}}$ for $\|\mathcal{F}\|_{\mathrm{op}}$ in (17). The left density obeys

$$\left\|q(DD^*)^r - q(a^2 I - B_0^2)\right\|_{\mathrm{op}} \le q\|E_{\mathrm{alg}}\|_{\mathrm{op}} + 2qa\|B - B_0\|_{\mathrm{op}}. \tag{19}$$

Proof. Singular-value decomposition gives $D(D^*D)^r = (DD^*)^r D$. Differentiate $B = D'D^{-1}$, substitute the definition of S, and subtract $r(B^2 - a^2 I)$. This gives (18) without commuting two different matrices. For (19), substitute the definition of E_{alg} and use $\left\|B^2 - B_0^2\right\|_{\mathrm{op}} \le (\|B\|_{\mathrm{op}} + \|B_0\|_{\mathrm{op}}) \|B - B_0\|_{\mathrm{op}} \le 2a \|B - B_0\|_{\mathrm{op}}$. □

The physical right density $q(D^*D)^r$ is not the same matrix as the left density in (19). Polar conjugation relates them, but that conjugating unitary may depend on x. To claim a fixed physical soliton basis one must also control this gauge, or invoke the exact commutation and fixed-space argument in the source. We do neither silently. In particular a small compressed Riccati residual is not a full physical equality certificate.

6 Worked examples and necessary costs

Example 6.1 (An exact rational Young test)**.** At $\gamma = 1$, $p = 3/2, r = 2, q = 3$, and $D_\sigma = 2/(3\sqrt{3})$. If $S, T \ge 0$, put $W = 3S^2$, $Z = T$. Then the deficit is the rational polynomial

$$Y = 2 \operatorname{tr} S^3 - 3 \operatorname{tr}(S^2 T) + \operatorname{tr} T^3. \tag{20}$$

For $\|S\|_{\mathrm{op}}, \|T\|_{\mathrm{op}} \le L$, Lemma 2.1 gives

$$Y \ge \frac{3}{4L} \left\|S^2 - T^2\right\|_{\mathrm{HS}}^2.$$

There is no assumption that S and T commute. For instance $S = \operatorname{diag}(1, 2)$ and $T = \left(\begin{smallmatrix} 1 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right)$ have $L = 2$; direct multiplication gives $Y = 13/2$ and $\left\|S^2 - T^2\right\|_{\mathrm{HS}}^2 = 77/8$. Thus the right side is $231/64 < 13/2$. Making these fields constant on an interval of length one and zero elsewhere gives measurable $L^{3/2}$ potentials and densities to which Theorem 2.2 applies. This is not an equality potential or a computed eigenvalue experiment.

Example 6.2 (The right-space rank cost is necessary)**.** Let $A(x) = s(\cos(\omega x), \sin(\omega x))$, $B = 0$, and $s, \omega > 0$. The physical row projector rotates with $\|Q'\|_{\mathrm{op}} = \omega$, whereas $\|A' - BA\|_{\mathrm{op}} = s\omega$. Thus the factor s^{-1} in the derivative bound is exact in this model. Vanishing residual as $s \downarrow 0$ does not force small channel rotation.

Example 6.3 (Small algebraic defect can coexist with rotation)**.** Let $u = (\cos(\omega x), \sin(\omega x))^T$, $v = (-\sin(\omega x), \cos(\omega x))^T$, and set

$$A = su, \quad B = \omega(uv^* + vu^*), \quad K^2 = (\omega^2 + s^{2r}) I_2.$$

Then $A' = BA$, $B^2 = \omega^2 I_2$, $[B, K^2] = 0$, and

$$F = s^{2r}(I_2 - uu^*), \quad \Phi = \min(1, s^{4r}), \quad \left\|P'\right\|_{\mathrm{op}} = \omega.$$

For bounded ω and $s \le 1$, a common finite κ is available. The action integrand tends to zero with s although the left space continues rotating. The fixture has no source connecting-path endpoints or trial orthonormality and does not contradict the source theorem. It rules out a rank-free deduction from the local retained defects alone.

Even exact active profiles have poor global rank floors. For the scalar solution $B = -a \tanh(ra(x - y))$, an exact frame is

$$A = a^{1/r} \operatorname{sech}^{1/r}(ra(x - y)).$$

On $|x - y| \le L$ its minimum is $s = a^{1/r} \operatorname{sech}^{1/r}(raL)$; this tends to zero exponentially as $L \to \infty$. Local rank-stable bounds therefore do not automatically assemble into a uniform whole-line stability theorem.

7 Trial forms and certified Gram normalization

The source approximates finitely many negative eigenfunctions in H^1, then orthonormalizes their L^2 Gram matrix. This is a form-domain argument, not a claim of operator-domain residual convergence. The following makes the finite normalization error explicit.

Proposition 7.1 (A common spectral-shift budget)**.** *Let $J : \mathbb{C}^n \to H^1(\mathbb{R};\mathbb{C}^m)$ be a trial synthesis, $G = J^*J$, and $F = (h_W[Je_i, Je_j])_{ij}$. Let $K_0 = \operatorname{diag}(k_1,\dots,k_n) > 0$, $\|K_0\|_{\mathrm{op}} \le \kappa_0$. Suppose certified operator-norm bounds give*

$$\|G - I\|_{\mathrm{op}} \le \varepsilon < 1, \qquad \left\|F + K_0^2\right\|_{\mathrm{op}} \le \eta.$$

The orthonormalized form $\widehat{F} = G^{-1/2} F G^{-1/2}$ satisfies

$$\left\|\widehat{F} + K_0^2\right\|_{\mathrm{op}} \le \frac{\eta + \kappa_0^2 \varepsilon}{1-\varepsilon}. \tag{21}$$

Hence any common ξ greater than or equal to this bound and less than $\min_i k_i^2$ gives $\widehat{F} \le -(K_0^2 - \xi I)$.

Proof. Put $S = G^{-1/2}$ and $t = (1-\varepsilon)^{-1/2}$. Functional calculus gives $\|S\| \le t$ and $\|S - I\| \le t - 1$. Since

$$\widehat{F} + K_0^2 = S(F + K_0^2)S + K_0^2 - SK_0^2S,$$

the norm is at most $\eta t^2 + \kappa_0^2(t-1)(1+t)$, which equals the right side of (21). This proof uses no commutation between G and K_0. The form domain is a linear space, so the finite coordinate change does not alter admissibility. The matrix order follows from its operator-norm bound. □

For example $\varepsilon = 1/100$, $\eta = 1/1000$, $\kappa_0^2 = 4$ give the bound $41/990 < 1/20$. If the smallest k_i^2 is at least one, the common shift $\xi = 1/20$ is valid. A finite trial inequality can prove variational upper bounds for negative eigenvalues. It does not exclude additional negative eigenvalues, establish equality in the full sum, or supply a fixed physical basis.

8 Exact profiles and the source proof dependencies

The converse profile calculation can be checked without reproducing the source's necessity argument. For $a > 0$, put

$$w = a\tanh(ra(x-y)), \qquad Q_a = \partial_x + w, \qquad D(Q_a) = H^1(\mathbb{R}).$$

Since w, w' are bounded, $Q_a^* = -\partial_x + w$ has domain H^1, and both second-order products have domain H^2. Direct multiplication gives

$$\begin{aligned} Q_a^*Q_a &= -\partial_x^2 + a^2 - (r+1)a^2\operatorname{sech}^2(ra(x-y)),\\ Q_aQ_a^* &= -\partial_x^2 + a^2 + (r-1)a^2\operatorname{sech}^2(ra(x-y)) \ge a^2. \end{aligned}$$

The L^2 kernel of Q_a is spanned by $\operatorname{sech}^{1/r}(ra(x-y))$. If $Q_a^*Q_a$ had an eigenvalue $\lambda \in (0, a^2)$, applying Q_a to its eigenfunction would give a nonzero eigenfunction of $Q_aQ_a^*$ with the same eigenvalue, a contradiction. The domain assertion follows either from these product domains and the eigenvalue equation, or from the polar decomposition of the closed operator Q_a. Thus the scalar channel has exactly one negative eigenvalue. With $t = \tanh(ra(x-y))$,

$$\int_{\mathbb{R}} ((r+1)a^2\operatorname{sech}^2(ra(x-y)))^p\,dx = \frac{(r+1)^p}{r}a^{2p-1}b_\sigma, \qquad D_\sigma\frac{(r+1)^p}{r} = 1.$$

This verifies equality for each stated profile and their constant unitary sums. It does not classify all equality cases.

The following map records the complete route in the cached source; having read it is not an independent certification of its global proof.

1. *Preliminaries and spectral bound.* The form construction, matrix Young inequality and its coercive variant precede a finite list of eigenfunctions. Smooth compact approximations are orthonormalized in the physical Gram matrix. A common downward shift preserves ordering and energy gaps. The real proof is transferred to complex Hermitian potentials by realification, which doubles both spectral multiplicities and the potential trace. Dividing by two is essential.

2. *Action and trace appendix.* For orthonormal rows U, a lower-triangular matrix C and a Hermitian connecting path B give $A = CU$, with endpoint values $K, -K$. The retained cost is $\|A' - BA\|_{\mathrm{HS}}^2 + c_*\Phi$. The trace appendix studies the full symbol $(sI - B)^2 + V + tI$; positivity of this symbol, not positivity of V alone, is the hypothesis. Half-line covariant minimizers give two boundary matrices. Their average is the inverse integrated resolvent, while their difference produces the nonnegative trace remainder. Order inside matrix products is retained throughout.

3. *Path appendix.* A regularized concave field and a penalized differential equation produce the connecting path through a compact zero-set and parity argument modulo row signs. The limiting procedure fixes the regularization before removing the penalty. A dimension-independent truncated Φ and its coefficient are already features of this source, not innovations of the present report. A finite Gram calculation cannot replace this connecting-path existence theorem.

4. *Compactness.* For equality the source obtains

$$T_j + \int_{\mathbb{R}} \{Y(W, Z_j) + \|A_j' - B_j A_j\|_{\mathrm{HS}}^2 + c_*\Phi_j\} \le d_j \longrightarrow 0.$$

   Here $T_j \ge 0$ suppresses mixing of distinct binding energies. The fixed physical column dimension m, together with the spectral tail $s_h = \sup_{i>h} k_i^2$, controls the infinite row tail:

$$\|Q_h A_j\|_{\mathrm{HS}}^2 \le m(s_h + e_j)^{1/r}, \qquad \|B_j Q_h\|_{\mathrm{op}} \le (s_h + e_j)^{1/2}.$$

   In this display Q_h is the source's row-tail projector, not the physical projector $Q = A^\dagger A$ used above. The algebraic error $e_j \to 0$ is taken on the selected full-measure set. Neither a finite row truncation nor the rank floor in our local theorem supplies these infinite-row compactness bounds.

5. *Rigidity.* The exact limit has fixed physical right spaces, orthogonal between different energies; there are at most m active directions. On each active space the exact algebraic commutation makes the left space fixed as well. Constant bases reduce the equation to an invertible square block and then to its Riccati equation. Complex realification is finally restricted back to the physical complex block; only identical scalar profiles can mix freely. This exact global argument remains an imported source result.

The report's reconstruction theorem improves the potential identification consequence of this budget, while its drift and Cayley theorems quantify selected finite-interval rigidity steps. They do not manufacture the source budget from the global spectral deficit for arbitrary near-extremizers.

9 Original Ram comparison and the physical residual

Three original chapters of *RamCalBooks*, Volume 21, were freshly read in full as text [6, 7, 8]. They provide specific safeguards rather than an alternative global Lieb–Thirring theorem.

The state-and-domain chapter requires domain transport for any operator similarity and distinguishes form norms from graph norms. This is directly relevant to Example 1.1: algebraic coefficients do not create an L^2 differential residual. For a self-adjoint H, the physical-metric chapter proves, with an injective synthesis $J : \mathbb{C}^n \to D(H)$, the identities

$$G = J^*J, \quad P_J = JG^{-1}J^*, \quad B_J = G^{-1}J^*HJ, \quad \|(H-\lambda)Jc\|^2 = \|(B_J-\lambda)c\|_G^2 + \|(I-P_J)HJc\|^2.$$

Indeed $P_J(H-\lambda)Jc = J(B_J-\lambda)c$, and the two orthogonal components give Pythagoras. Hermiticity of B_J means G-Hermiticity, not Euclidean Hermiticity. If only $J : \mathbb{C}^n \to H^1$ is known, the compressed form still exists but the displayed operator residual need not. This distinction is preserved in Proposition 7.1. No loss of the unretained component is allowed.

The spectral-limits chapter supplies a concrete moving-vector pollution example and separates strong-resolvent statements from uniform residual bounds. It also gives the appropriate positive semibounded Rayleigh approximation under form-density hypotheses. We do not import its indefinite example as a claim that every semibounded Galerkin scheme suffers the same pathology. Here its relevant lesson is narrower: a finite negative Ritz list need not exhaust the negative spectrum or certify equality of its full Riesz sum.

The new synthesis is therefore physical and quantitative: use the source's actual retained Young/action errors, reconstruct the physical potential in L^p, track range projectors in their correct spaces, and report the cost of rank, energy separation and endpoint distance. Trace convexity, projector differentiation, Cauchy–Schwarz and Cayley linearization are classical tools. Neither their names nor the use of a Gram matrix is credited as a new mathematical principle.

10 Current public context and evidence boundaries

The public refresh was performed on October 9, 2026. Read–Schulz [2], version 2 of September 14, state the sharp exponent-one constant and an operator-valued extension; their abstract is not a statement of the whole interval of exponents and matrix equality classification studied here. Frank–Gontier–Lewin [3] distinguish finite-rank optimizers and, at the endpoint $\gamma = 3/2$ in one dimension, the KdV multi-soliton family. That endpoint result must not be identified with the interior one-state-per-channel formula (3). Classical matrix factorization has earlier context in Benguria–Loss [4]. The first and third external papers were checked at primary abstract/version level; the second also at its opening exposition. Their full proofs were not independently audited here.

The publisher's family-262 documentation [5] was also read. Its selected comparator statement describes a scalar sharp constant and attainment problem. That scope description is not evidence that the matrix equality classification was executed or verified in a proof assistant. No Lean computation was performed. The cached October 5 manuscript is analyzed as a source claim, not converted into an accepted theorem merely by its repository location or by successful local tests.

All 12 cached public text files were literally displayed and read: 2,226 lines, including all eight scientific modules (2,061 lines). The exact per-file intervals and before/after hashes are recorded in the owned reading ledger. The three private original chapters contain 616 lines, all freshly displayed; their separately included figures and other cited private documents were not inspected. There was no whole-corpus audit. The complete text read does not assert an independent verification of the

path topology, all functional-analytic limits, or the global equality classification. The parent source identity record concerns two captured trees, not current live-branch status.

An independently executable owned checker tests rational noncommuting Young fixtures, projector and commutator identities, the ordered Cayley forcing equation, second-order forcing, common-shift normalization and scalar profile identities. Its receipt separates parameter cases from assertions. These are supplemental algebraic regressions, not quadrature-based claims about arbitrary measurable potentials, and no other report's or reviewer's test counts are included.

11 Advantages, costs and future work

The potential reconstruction estimate requires no pointwise cap and no additional uniform integrability hypothesis: the actual source coercivity pays for its weight. It gives a global $L^p(\mathrm{HS})$ bound while channel stability is deliberately local. The truncated-defect argument treats the entire interval, including the bad set, and has no extra row-dimension factor. The Cayley estimate retains noncommuting forcing and replaces a growing forward Gronwall factor by a decaying kernel on its stated spectral chart. These are useful refinements of different steps, not three substitutes for the same missing global hypothesis.

The costs are explicit. The reconstruction constant depends on p and total potential mass. Left-space drift can deteriorate like s^{-2r}; physical energy separation costs g^{-1}; the Riccati chart costs η^{-2}. Exact soliton tails force some of these quantities to degenerate. No numerical conditioning advantage or computational speedup has been measured. The source coefficient c_* must be available to turn the symbolic action budget into a numerical one.

A genuine near-equality theorem would need a construction that turns the spectral deficit into compatible global action budgets, controls tail escape and the number of active channels, and patches rank-stable intervals without accumulating uncontrolled gauge rotation. The two Riccati charts suggest a local patching method, but overlap alignment and physical polar factors require separate estimates. Repeated or nearly coincident energies need cluster projectors rather than artificially separated channels. These are concrete future obligations, not conclusions of the finite certificates proved here.

In summary, C3 denotes notable source-specific quantitative reconstruction and conditional rigidity estimates. It does not denote a new historical priority claim, an independent global classification, or a change in the catalogue's L3 applicability. No source correction is asserted: the report retains the source's valid truncation, fixed-column compactness, common-shift and constant-basis safeguards, and strengthens selected consequences.

References

[1] OpenAI, *Equality cases in the sharp one-dimensional matrix Lieb–Thirring inequality*, cached manuscript, October 5, 2026. Public repository, https://github.com/openai/math. All twelve cached text files read; global conclusions attributed to the manuscript.

[2] L. Read and M. R. Schulz, *The sharp one-dimensional Lieb–Thirring inequality for the sum of eigenvalues*, arXiv:2609.10478v2, September 14, 2026. https://arxiv.org/abs/2609.10478v2.

[3] R. L. Frank, D. Gontier and M. Lewin, *Optimizers for the finite-rank Lieb–Thirring inequality*, arXiv:2109.05984v2, May 11, 2023. https://arxiv.org/abs/2109.05984.

[4] R. Benguria and M. Loss, *A simple proof of a theorem of Laptev and Weidl*, arXiv:math-ph/9906024v2, September 30, 1999. https://arxiv.org/abs/math-ph/9906024.

[5] OpenAI, *Family 262 scope documentation*, accessed October 9, 2026. https://github.com/openai/math/blob/main/lean/docs/262.md. Metadata read; no proof-assistant execution.

[6] M. Subbarao, *RamCalBooks*, Volume 21, original chapter `ch01a-state-domain.tex`, 164 lines. Confidential local proof source, freshly read in full.

[7] M. Subbarao, *RamCalBooks*, Volume 21, original chapter `ch01d-physical-metrics.tex`, 210 lines. Confidential local proof source, freshly read in full.

[8] M. Subbarao, *RamCalBooks*, Volume 21, original chapter `ch01f-spectral-limits.tex`, 242 lines. Confidential local proof source, freshly read in full.